

Lecture 7 - Actions and Labels

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- *Updates* are actions which modify a stateful variable.
- *Intereference* occurs when updates are performed arbitrarily, potentially leading to unexpected behaviour and race conditions.
- *Mutual exclusion* is the process of preventing two concurrently running process from being able to modify data at the same time to prevent interference.
 - Mutual exclusion is implemented using *locks*.
 - * When a process wants to be able to read/write to data, it must request a lock for that data. If no other process has a lock on the data, the lock is acquired by the process, giving it exclusive access to that data until the lock is released.
- An *active* process is a process which initiates actions.
- A *passive* process is a process which responds to actions.
- A *semaphore* is a variable used to implement synchronization between concurrently running processes.
- A process is *awakened* if **I DON'T FULLY UNDERSTAND WHAT AWAKENING IN THIS CONTEXT MEANS**
 - Semaphores can only take on non-negative integer values.
- Only the following operations can be performed on a given semaphore **s**.
 - **up(s)**
 - * If a process is blocked on **s**, then the process is awakened, otherwise, **s** is awakened.
 - **down(s)**
 - * If **s** > 0, the **s** is decremented, otherwise, the process calling **down(s)** is blocked.
- *Binary semaphores* are a special type of semaphore which can only take on the values of 0 or 1.
 - Binary semaphores are used to implement locks on data.